पेटेंट कार्यालय
शासकीय जर्नल

OFFICIAL JOURNAL
OF
THE PATENT OFFICE

निर्गमन सं. 12/2026	शुक्रवार	दिनांक: 20/03/2026
ISSUE NO. 12/2026	FRIDAY	DATE: 20/03/2026

पेटेंट कार्यालय का एक प्रकाशन
PUBLICATION OF THE PATENT OFFICE

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/02/2026
(21) Application No.202621011703 A
(43) Publication Date : 20/03/2026

(54) Title of the invention : INTELLIGENT SPAM EMAIL DETECTION SYSTEM USING MACHINE LEARNING WITH TF-IDF BASED FEATURE EXTRACTION

(51) International classification	:H04L 12/58, G06N 20/00, G06N 20/10, G06F 16/35, G06Q 10/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

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(57) Abstract :
ABSTRACT INTELLIGENT SPAM EMAIL DETECTION SYSTEM USING MACHINE LEARNING WITH TF-IDF BASED FEATURE EXTRACTION The present invention discloses an intelligent spam email detection system (100) utilizing machine learning algorithms with Term Frequency-Inverse Document Frequency (TF-IDF) based feature extraction. The system comprises a data input module (110) for receiving emails, a text preprocessing module (120) implementing tokenization, stop-word removal, stemming, and lemmatization, and a feature extraction module (130) generating TF-IDF weighted feature vectors. A machine learning classification module (140) implements supervised learning classifiers including Naive Bayes, Support Vector Machine, Logistic Regression, and Random Forest for binary classification of emails as spam or legitimate. A model training engine (150) facilitates classifier training using labeled datasets with cross-validation. The output classification module (160) generates classification results with confidence scores. The system achieves 96.2 percent classification accuracy with the Naive Bayes classifier, providing effective spam filtering while minimizing false positives and adapting to evolving spam patterns.

No. of Pages : 20 No. of Claims : 10